Ordinary Annuities

$$\boxed{Future\ Value\ =\ PMT\left(\frac{(1+\frac{r}{n})^{nt}-1}{\frac{r}{n}}\right)}$$



1st Payment 2nd Payment (nt-1)th Payment Last Payment

$$Future\ Value\ = PMT(1+\frac{r}{n})^{nt-1} + PMT(1+\frac{r}{n})^{nt-2} + \cdots + PMT(1+\frac{r}{n})^{1} + PMT$$

$$= PMT\left(\frac{(1+\frac{r}{n})^{nt}-1}{(1+\frac{r}{n})-1}\right)$$ Sum of a Geometric Sequence

$$= PMT\left(\frac{(1+\frac{r}{n})^{nt}-1}{\frac{r}{n}}\right)$$

$$\boxed{Present\ Value\ = PMT\left(\frac{1-(1+\frac{r}{n})^{-nt}}{\frac{r}{n}}\right)}$$



$$Present\ Value(1+\frac{r}{n})^{nt} = PMT\left(\frac{(1+\frac{r}{n})^{nt}-1}{\frac{r}{n}}\right)$$

$$Present\ Value\ = \frac{PMT\left(\frac{(1+\frac{r}{n})^{nt}-1}{\frac{r}{n}}\right)}{(1+\frac{r}{n})^{nt}} = PMT\left(\frac{1-(1+\frac{r}{n})^{-nt}}{\frac{r}{n}}\right)$$